

Homework 2: Diagonalization

M. Usman Rashid

BCS A, E, F, G

Problem 1. Consider the linear transformation $L(A) = A^T$ from $\mathbb{R}^{2 \times 2}$ to $\mathbb{R}^{2 \times 2}$. Is transformation L diagonalizable? If so, find an eigenbasis for L . [Hint: Consider symmetric and skew-symmetric matrices.]

Problem 2. Consider the linear transformation $D(f) = f'$ (the derivative) from C^∞ to C^∞ . Show that all real numbers are eigenvalues of D . [Hint: Apply D to exponential functions.]

Problem 3. Diagonalize matrix A if you can.

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}.$$

Problem 4. Diagonalize matrix A if you can.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Problem 5. Diagonalize matrix A if you can.

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}.$$

Submission Date: September 29, 2024 (11:59PM)

Good Luck